

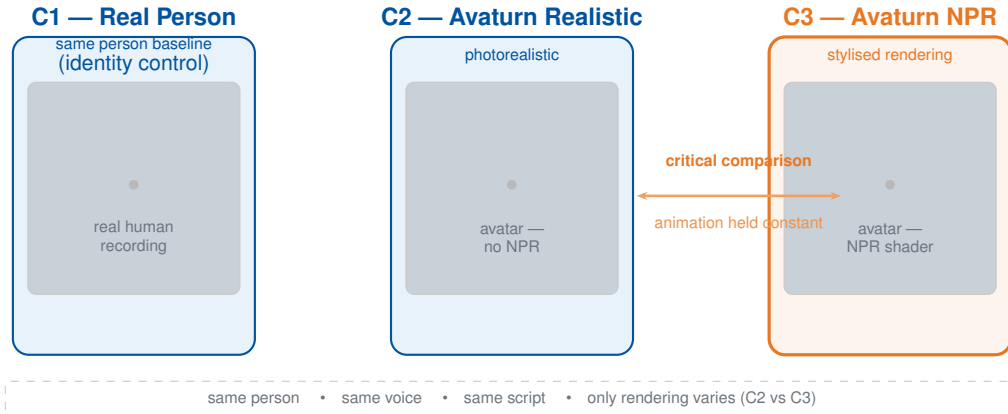
# User Study

## Perception of Virtual Avatar Advisors

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# Three Study Conditions



# Three Advisor Scenarios

| Scenario                                               | Context                                                                                                              | Option A                                                                                                                                                             | Option B                                                                                                                                                         | Recommends                                                        |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>S1</b><br>Career Choice<br>Competence               | Master's degree just completed. Two job offers, one month start. No financial pressure. Five days to decide.         | <b>Job A</b><br>Generalist role, small company. Broad scope, high individual visibility. Skills transfer across industries. <b>Salary approx. 8 000 EUR/yr less.</b> | <b>Job B</b><br>Specialist role, large well-known company. Higher salary, recognised brand. Deep specialisation within a large team, less individual visibility. | <b>Job A</b><br><i>"Pivot option won't come back later"</i>       |
| <b>S2</b><br>Travel Route<br>Benevolence<br>Competence | Arrived in unfamiliar city. Must reach hotel in one hour for a colleague meeting. Two public transport routes found. | <b>Route A</b><br>20 min, one transfer at an unfamiliar station. Arrives 40 min early <i>if the connection holds.</i>                                                | <b>Route B</b><br>50 min, one direct line, no transfers. Arrives 10 min before meeting. <b>Reliable.</b>                                                         | <b>Route B</b><br><i>"Certainty beats a buffer at risk"</i>       |
| <b>S3</b><br>Learning Resource<br>Integrity            | Wants to learn data analysis for CV. A few weeks available, two options found. Must commit to one.                   | <b>Option A</b><br>Structured course, 35 EUR, 8 hrs. <b>CV certificate from named platform.</b>                                                                      | <b>Option B</b><br>Free tutorial series by a practitioner. Same content, flexible pace. No certificate.                                                          | <b>Option A</b><br><i>"Certificate signals the skill on a CV"</i> |

# What the Avatar Says — Recommendation Scripts

**S1**

Career Choice

→ Job A

*“Based on what you’ve shared, I’d go with Job A. Yes, the salary is lower right now, but the generalist role gives you experience breadth that is genuinely hard to get later. Once you specialise, switching back is costly. Take the pivot while you still can.”*

**S2**

Travel Route

→ Route B

*“I’d take Route B. Route A sounds faster on paper, but you are in an unfamiliar city and that transfer is a gamble. You have a colleague meeting and missing it is not an option. Route B gets you there with time to spare. Take the reliable one.”*

**S3**

Learning Resource

→ Option A

*“I would go with the paid course. The certificate matters here. It is not just about what you learn, it is about what you can show. A recruiter cannot verify a free tutorial series. Thirty-five euros for something you can put on your CV is worth it.”*

# Video Production Checklist (researcher note — before recording)

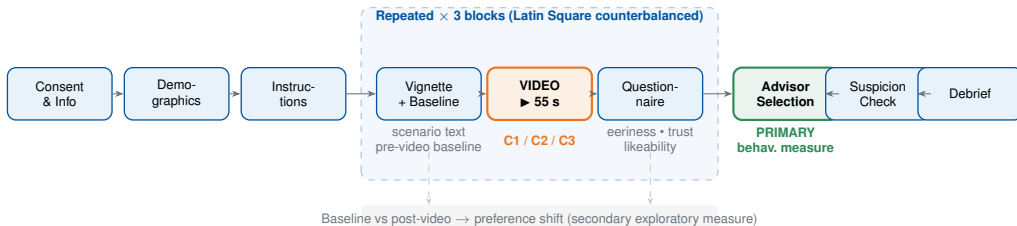
## Match across C1, C2, C3

- ▶ **Resolution & ratio:** 1920×1080, 16:9 for all three
- ▶ **Frame rate:** 30 fps — Unity export must match camera
- ▶ **Duration:** within  $\pm 1$  s per scenario
- ▶ **Framing:** eye-level, centred, same head-to-shoulder crop
- ▶ **Background:** same desk, setting, camera angle replicated in Unity scene

## Audio & technical

- ▶ **Audio:** one recording, same mic and room, applied to all three; normalise to -16 LUFS
- ▶ **No TTS** in any condition
- ▶ **Lighting colour:** match Unity light temperature to C1 camera environment
- ▶ **Codec:** H.264 MP4, same bitrate for all three

# Study Flow Overview



## Scene 1 — Welcome & Consent

Advisor Evaluation Study

### Welcome

This study examines how people evaluate advice from different types of advisors. You will watch three short videos and answer questions about your impressions.

**Duration:** approx. 20–25 min

**Data:** fully anonymous

**Device:** laptop / desktop

☐ I have read the above information and agree to participate.

[Start Study >](#)

## Scene 2 — Participant Instruction

Advisor Evaluation Study

Please complete this study on a **laptop or desktop computer**. A phone screen is too small for accurate viewing. **If you are currently on a mobile phone or tablet, please switch devices before continuing.**

You will watch **three short videos**. Each video shows a virtual advisor making a recommendation. After each video, you will answer a set of questions about your immediate impression of the advisor.

**There are no right or wrong answers.** Please respond based on your honest first impression, not what you think the “correct” answer is.

Please watch each video in full before answering.

Continue >



## Scene 2b — Demographics (1 of 2)

Advisor Evaluation Study

About You — Basic InformationPart 1 of 2

**Q1.** What is your age?

enter years

**Q2.** What is your gender?

☐ Man☐ Woman☐ Non-binary

☐ Prefer not to say☐ Other: \_\_\_\_\_

**Q3.** What is your highest completed level of education?

☐ Secondary school☐ Bachelor's degree☐ Master's degree

☐ Doctoral degree☐ Other: \_\_\_\_\_

Continue >

## Scene 2c — Demographics (2 of 2)

Advisor Evaluation Study

Avatar Experience & Trust Baseline

Part 2 of 2

**Q4.** How often do you interact with virtual avatars or digital characters?

☐ Never

☐ Rarely (few times/year)

☐ Occasionally (monthly)

☐ Regularly (weekly)

☐ Daily

**Q5.** In general, I tend to trust people I meet for the first time.

Propensity to trust covariate — Mayer et al. (1995)

1 = Strongly Disagree

4 = Neutral

7 = Strongly Agree

1

2

3

4

5

6

7

Continue >

## Scene 3 — Scenario Page (Career Choice)

Scenario 1 of 3 — Before You Watch

### Scenario Context (read before watching)

You recently completed your master's degree and have received two job offers in your field. Both positions start in one month. You have no strong financial pressure and both cities are equally acceptable to you. You have five days to decide.

#### Job A

A broad generalist role at a smaller company. The work area is yours to define, the team is small, and your contributions are directly visible. Skills transfer across industries. **Starting salary approx. 8 000 EUR/yr less than Job B.**

#### Job B

A specialist role at a large, well-known company in a high-demand field. The salary is substantially higher and the employer name carries weight on a CV. The role involves deep specialisation within a large team, with less individual visibility.

### Before watching — which option would you initially lean toward?

☐ Job A☐ Job B☐ No clear preference

Your answer is saved before the video plays — you cannot go back to change it.

Continue to Video >

↓ scroll to review options again before answering

## Scene 3b — Scenario Page (Travel Route)

Scenario 2 of 3 — Before You Watch

### Scenario Context (read before watching)

You have just arrived by train in a city you have never visited before. You need to reach your hotel, where you are meeting a group of colleagues in about one hour. You have checked a map app and found two routes, both using public transport.

#### Route A

Takes around 20 minutes with one transfer at a mid-size station you have never used. If the connection runs on time, you arrive approximately **40 minutes early**.

#### Route B

Takes around 50 minutes on a single direct line with no transfers. It loops through the city centre. You arrive approximately **10 minutes before** your meeting.

### Before watching — which option would you initially lean toward?

☐ Route A

☐ Route B

☐ No clear preference

Your answer is saved before the video plays — you cannot go back to change it.

Continue to Video >

↓ scroll to review options again before answering

## Scene 3c — Scenario Page (Learning Resource)

Scenario 3 of 3 — Before You Watch

Scenario Context (read before watching)

You want to learn data analysis to strengthen your CV. You have set aside a few weeks and found two options. You need to decide which one to commit to.

Option A

A structured online course from a well-known learning platform. Costs 35 euros, runs about eight hours over a few weeks, and **awards a certificate** you can add to your CV. The curriculum is set out from start to finish.

Option B

A free video tutorial series by a practitioner with a strong reputation and a large following. The content covers the same material at no cost. **No certificate**, and you set your own pace.

Before watching — which option would you initially lean toward?

☐ Option A

☐ Option B

☐ No clear preference

Your answer is saved before the video plays — you cannot go back to change it.

Continue to Video >

↓ scroll to review options again before answering

## Scene 4 — Block Structure & Part A Comprehension Check

This block repeats × 3. Videos are always labelled “Video 1 / 2 / 3” — condition names C1/C2/C3 are never shown.



### Part A. Video Comprehension Check (exclusion criterion: wrong answer or “did not watch” → exclude)

**A1.** What did the advisor in this video recommend?

- ☐ Option A
- ☐ Option B
- ☐ I did not watch the full video

*Answer options configured per scenario in SoSci Survey.*

## Scene 5 — Parts B & C: Likeability + Human-Likeness (Godspeed)

### Part B. Likeability — 5 items

1 2 3 4 5

**B1** Dislike ☐ ☐ ☐ ☐ ☐ Like

**B2** Unfriendly ☐ ☐ ☐ ☐ ☐ Friendly

**B3** Unkind ☐ ☐ ☐ ☐ ☐ Kind

**B4** Unpleasant ☐ ☐ ☐ ☐ ☐ Pleasant

**B5** Awful ☐ ☐ ☐ ☐ ☐ Nice

**B score:** mean(B1–B5). All 5 items verbatim from Godspeed III (Bartneck et al., 2009). No modifications.

### Part C. Human-Likeness — 5 items

1 2 3 4 5

**C1** Fake ☐ ☐ ☐ ☐ ☐ Natural

**C2** Machinelike ☐ ☐ ☐ ☐ ☐ Humanlike

**C3** Unconscious ☐ ☐ ☐ ☐ ☐ Conscious

**C4** Artificial ☐ ☐ ☐ ☐ ☐ Lifelike

**C5** Moving rigidly ☐ ☐ ☐ ☐ ☐ Moving elegantly

**C score:** mean(C1–C5). All 5 items verbatim from Godspeed I (Bartneck et al., 2009). C5 anchor is *Moving elegantly* (not “naturally”) — exact original wording.

## Scene 6 — Part D: Eeriness (Ho & MacDorman, 2010 — 8 items verbatim, 5-point)

| Rate your impression of the advisor |             | 1 = negative anchor   |                       |                       |                       |                       | 5 = positive anchor |
|-------------------------------------|-------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|---------------------|
|                                     |             | 1                     | 2                     | 3                     | 4                     | 5                     |                     |
| D1                                  | Reassuring  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Eerie               |
| D2                                  | Numbing     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Freaky              |
| D3                                  | Ordinary    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Supernatural        |
| D4                                  | Bland       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Uncanny             |
| <hr/>                               |             |                       |                       |                       |                       |                       |                     |
| D5                                  | Unemotional | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Hair-raising        |
| D6                                  | Uninspiring | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Spine-tingling      |
| D7                                  | Boring      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Shocking            |
| D8                                  | Predictable | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Thrilling           |



Scene 7 — Part E: Trust Competence (Canales et al., 2024 — 4 items, 7-point Likert)

| Part E. Trust: Competence — 4 items (7-point Likert, 1 = SD, 7 = SA)                                                                                                                                              |  |                       |                       |                       |                       |                       |                       |                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 1 = Strongly Disagree    4 = Neutral    7 = Strongly Agree                                                                                                                                                        |  | 1                     | 2                     | 3                     | 4                     | 5                     | 6                     | 7                     |
| E1. I believe this advisor is capable and competent.                                                                                                                                                              |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| E2. I would rely on advice given by this advisor.                                                                                                                                                                 |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| E3. This advisor seems knowledgeable.                                                                                                                                                                             |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| E4. This advisor seems to have the expertise needed to give sound advice on this kind of decision.                                                                                                                |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Source: Canales et al. (2024); McKnight &amp; Chervany (2001)</b>                                                                                                                                              |  |                       |                       |                       |                       |                       |                       |                       |
| E score: mean(E1–E4)    Higher = stronger perceived competence                                                                                                                                                    |  |                       |                       |                       |                       |                       |                       |                       |
| Scale: 1 = Strongly Disagree    4 = Neither agree nor disagree    7 = Strongly Agree                                                                                                                              |  |                       |                       |                       |                       |                       |                       |                       |
| Tests H2: Does NPR rendering (C3) produce significantly lower competence ratings than photorealistic avatar (C2)? Equivalence is predicted — self-reported competence trust should not differ by rendering style. |  |                       |                       |                       |                       |                       |                       |                       |

# Scene 8 — Part F: Trust Benevolence (Canales et al., 2024 — 4 items, 7-point Likert)

| Part F. Trust: Benevolence — 4 items (7-point Likert, 1 = SD, 7 = SA)                           |  |  |                       |                       |                       |                       |                       |                       |                       |
|-------------------------------------------------------------------------------------------------|--|--|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 1 = Strongly Disagree    4 = Neutral    7 = Strongly Agree                                      |  |  | 1                     | 2                     | 3                     | 4                     | 5                     | 6                     | 7                     |
| F1. I feel comfortable in the presence of this advisor.                                         |  |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| F2. This advisor appears to have good intentions.                                               |  |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| F3. I believe this advisor would act in my best interest.                                       |  |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| F4. I felt this advisor was looking out for my best interest, not just giving a generic answer. |  |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Source: Canales et al. (2024); McKnight &amp; Chervany (2001)</b>                            |  |  |                       |                       |                       |                       |                       |                       |                       |
| F score: mean(F1–F4)    Benevolence = warmth, caring, acting in the participant's interest      |  |  |                       |                       |                       |                       |                       |                       |                       |
| Scale: 1 = Strongly Disagree    4 = Neither agree nor disagree    7 = Strongly Agree            |  |  |                       |                       |                       |                       |                       |                       |                       |
| Note: Benevolence may be most sensitive to eeriness via affective social inference.             |  |  |                       |                       |                       |                       |                       |                       |                       |

# Scene 9 — Part G: Trust Integrity (Canales et al., 2024 — 3 items, 7-point Likert)

## Part G. Trust: Integrity — 3 items (7-point Likert, 1 = SD, 7 = SA)

1 = Strongly Disagree    4 = Neutral    7 = Strongly Agree

1    2    3    4    5    6    7

**G1.** This advisor reasons transparently, acknowledging trade-offs rather than oversimplifying.

☐ ☐ ☐ ☐ ☐ ☐ ☐

**G2.** This advisor appears to be honest.

☐ ☐ ☐ ☐ ☐ ☐ ☐

**G3.** I believe this advisor would keep its word.

☐ ☐ ☐ ☐ ☐ ☐ ☐

**Source: Canales et al. (2024); McKnight & Chervany (2001)**

**G score:** mean(G1–G3)    Higher = stronger perceived integrity

Scale: 1 = Strongly Disagree    4 = Neither agree nor disagree    7 = Strongly Agree

*Integrity captures transparency and intellectual honesty. G1 directly tests whether the advisor acknowledges trade-offs rather than oversimplifying recommendations.*

## Scene 10 — Part H: Recommendation Response + Attention Check

### Part H. Response to the Recommendation — 1 item (7-point Likert)

1 = Strongly Disagree   4 = Neutral   7 = Strongly Agree

1   2   3   4   5   6   7

**H1.** How much do you agree with the recommendation this advisor gave?

☐ ☐ ☐ ☐ ☐ ☐ ☐

**H1** measures recommendation congruence: did the advisor's reasoning shift the participant's preference? Used for the within-block preference-shift analysis (compare pre-video baseline to H1). Not compared across conditions directly (scenario content confound). H2 (warmth) was removed as redundant with Part B Likeability.

### AC. Attention Check — embedded once per block

“To confirm you are reading carefully, please select option **[N]** regardless of anything else.”  
Target value varies per block:

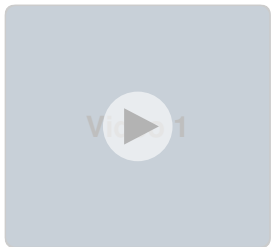
**Block 1: select 3   Block 2: select 4   Block 3: select 2**

Participants who fail any one check are excluded from all analyses.

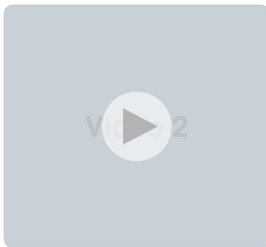
## Scene 11 — Post-Study: Advisor Preference Ranking (P1)

### ★ P1. Primary Behavioural Measure (adapted from Canales et al., 2024)

Imagine each advisor could give you advice on any type of decision. **Order** them from the one you would **most** want to consult (1) to the one you would **least** want to consult (3).



Rank: \_\_\_\_\_



Rank: \_\_\_\_\_



Rank: \_\_\_\_\_

In SoSci Survey: drag-and-drop rank order question. Advisor labels show as "Video 1 / 2 / 3" — condition names (C1/C2/C3) are never revealed to participants. Latin Square group mapping applied during data cleaning.

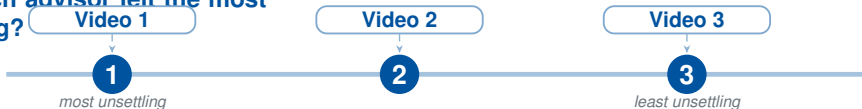
## Scene 12 — Comparative Preference Ranking (P2, P3, P4)

Post-Study — drag advisor chips to desired rank position

**P2. Which advisor felt the most trustworthy?**



**P3. Which advisor felt the most unsettling?**



**P4. Which advisor is most suitable for a professional context?**



## Post-Study — Open Reflection (P5, P6, J)

### Post-Study — Open Reflection

**P5. Looking back at all three advisors, what was the most important factor in deciding which one you trusted the most?**

open text response

**P6. Did the visual appearance of any advisor distract you from the content of the recommendation? If yes, describe.**

open text response

**J. Was there a specific moment, comment, or visual detail that noticeably increased or decreased your trust? (*Block 2 only*)**

open text response

### Responses used for exploratory qualitative analysis

P5 and P6 are open-ended and optional. J appears in Block 2 only. Responses supplement the quantitative scales.

Responses will be analysed thematically and reported alongside quantitative findings. Optional — participants who skip are not excluded.

## Part I. Visual Style Check — Manipulation Check

Shown ONCE after all three videos. Per-block placement would prime participants to attend to visual style.

### I1. How visually stylised or non-realistic did each advisor appear?

Scale: 1 = Completely realistic 7 = Very stylised

Completely realistic (1)

1 2 3 4 5 6 7

I1a. Video 1 (Advisor 1)

☐ ☐ ☐ ☐ ☐ ☐ ☐

I1b. Video 2 (Advisor 2)

☐ ☐ ☐ ☐ ☐ ☐ ☐

I1c. Video 3 (Advisor 3)

☐ ☐ ☐ ☐ ☐ ☐ ☐

**I1 score:** one rating per advisor. Expected pattern: C3 (NPR) scores highest, C1 (real person) lowest. A significant difference confirms the rendering manipulation was perceived as intended.



### Part I. Visual Style Check — Manipulation Check (continued)

#### I2. The visual appearance of this advisor was noticeably artificial or non-realistic.

Scale: 1 = Strongly Disagree 7 = Strongly Agree

Strongly Disagree (1)

1 2 3 4 5 6 7

I2a. Video 1 (Advisor 1)

☐ ☐ ☐ ☐ ☐ ☐ ☐

I2b. Video 2 (Advisor 2)

☐ ☐ ☐ ☐ ☐ ☐ ☐

I2c. Video 3 (Advisor 3)

☐ ☐ ☐ ☐ ☐ ☐ ☐

**I2 score:** one rating per advisor. Complements I1 with an agree/disagree format. Together I1 and I2 confirm whether participants perceived the rendering difference between C2 (photorealistic) and C3 (NPR).

## Post-Study — Appearance Influence (HA)

### HA. Appearance Influence (rate all 3 advisors)

For each advisor, to what extent did their visual appearance influence how you felt about their recommendation?

1 = Not at all    4 = Somewhat    7 = Very strongly

1 = Not at all    4 = Somewhat    7 = Very strongly

**1   2   3   4   5   6   7**

HA1. Advisor from Video 1

☐ ☐ ☐ ☐ ☐ ☐ ☐

HA2. Advisor from Video 2

☐ ☐ ☐ ☐ ☐ ☐ ☐

HA3. Advisor from Video 3

☐ ☐ ☐ ☐ ☐ ☐ ☐

**HA score:** rated per advisor. High HA + lower trust in C3 supports a visual salience explanation. Low HA + lower trust suggests an automatic uncanny valley response not mediated by conscious attention to style.

